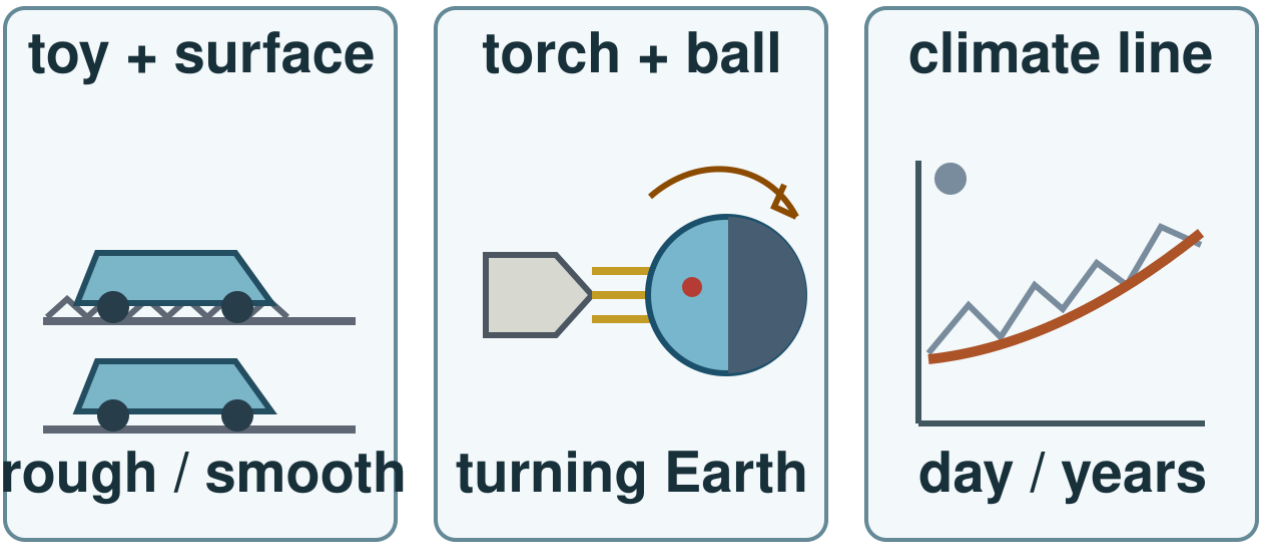
Knowledge organiser

I can match a question to a model.

I can give a claim with a matching clue.

I can say one thing the model cannot show.

I am learning to… choose a model that fits a science question and use a matching clue.



Choose a model: toy and surface, torch and ball, or long climate record. Sketches are simplified; the climate line is not data.

I can match a question to a model. I can give a claim with a matching clue. I can say one thing the model cannot show.

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Word page

I can match a question to a model.

I can give a claim with a matching clue.

I can say one thing the model cannot show.

Word	It means	Example
force	a push or pull on an object	Friction is a force that can slow a toy.
rotation	a turn around an axis	Earth’s rotation brings day and night.
evidence	an observation or result that supports a claim	The shorter run is evidence for my answer.
climate	weather patterns over many years	A hot day alone does not show a climate trend.

Read the question first. Choose the model that can show a useful clue. Give the clue, your claim and one limit.

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Arrival and starter

I can match a question to a model.

I can give a claim with a matching clue.

I can say one thing the model cannot show.

Match each question to a kit card: toy and surface, globe and torch, or warming chain.

Think back. Say, point or write one answer for each question.

1. Give one link in the warming chain.

2. Name one fact our Solar System cards support.

3. How can friction affect a moving toy?

You may point, sign, say or write your first ideas.

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I do · model and check

I can match a question to a model.

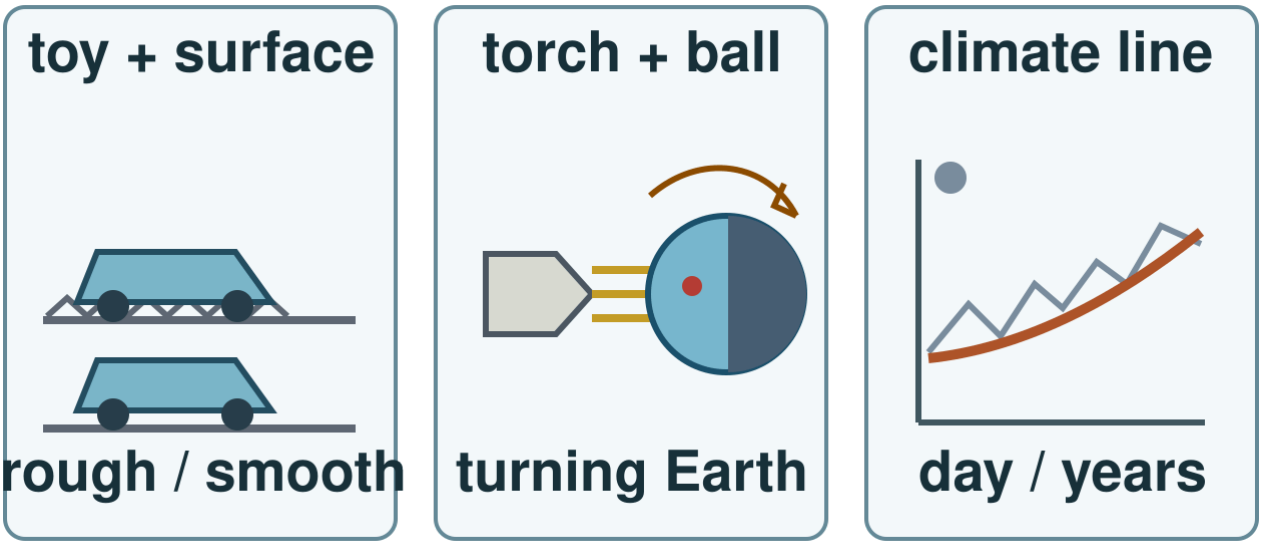
I can give a claim with a matching clue.

I can say one thing the model cannot show.

I choose a toy for a motion question. The toy went less far on rough paper in this run. I must repeat before I claim why.

Watch: Toy motion question → choose toy and surface → measure distance → test friction as one cause.

Show me: If my question is about day and night, should I point to the toy or torch-and-ball?



Choose a model: toy and surface, torch and ball, or long climate record. Sketches are simplified; the climate line is not data.

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We do · sort and explain

I can match a question to a model.

I can give a claim with a matching clue.

I can say one thing the model cannot show.

Torch and ball are chosen for day and night. Add the missing clue. Say why it fits.

Our turn: Day and night question → choose torch and ball → observe ____ → explain Earth's turn.

Talk, then sort.

Card	Choose a group
Why did the toy slow?	toy and surface / torch and ball / warming cards
Why is a place in night?	toy and surface / torch and ball / warming cards
Why is climate warming?	toy and surface / torch and ball / warming cards

Supported: Show the torch and marked ball; point to the place that moves into shade.

Independent: Give an answer and a matching clue.

Aspire: Explain why a toy run cannot test a climate claim even when it gives a clear result.

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We do 2 · choose and explain

I can match a question to a model.

I can give a claim with a matching clue.

I can say one thing the model cannot show.

What helps test a toy-motion claim?

A. A toy run with the same start

B. One day of weather

All of us: A model trolley stops sooner on a mat than on a smooth board. Which idea is relevant to test?

A. The length of Earth’s orbit.

B. Friction at the trolley surface.

C. The amount of daylight outside.

D. The climate of the whole Earth.

Show your choice and say why. You may use a point, card, board or sign. Take one quiet minute to reset.

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Independent · Supported

I can match a question to a model.

I can give a claim with a matching clue.

I can say one thing the model cannot show.

Choose two stations. Give a claim and matching clue for each. Name one limit.

Talk first. You can plan in any language.

Point, sign, speak, draw, type, write, use a scribe or show a staff-captured photo.

Match a question to a model.

Say first: For ___, I choose the ___ model.

Question or model	My claim and clue	One limit
_____	_____	_____
_____	_____	_____

Choose one route within the full 8 minutes. Keep your own response.

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Independent · Independent

I can match a question to a model.

I can give a claim with a matching clue.

I can say one thing the model cannot show.

Choose two stations. Give a claim and matching clue for each. Name one limit.

Talk first. You can plan in any language.

Point, sign, speak, draw, type, write, use a scribe or show a staff-captured photo.

Give an answer and a matching clue.

Say first: My claim is ___; the clue is ___.

Question or model	My claim and clue	One limit
_____	_____	_____
_____	_____	_____

Choose one route within the full 8 minutes. Keep your own response.

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Independent · Aspire

I can match a question to a model.

I can give a claim with a matching clue.

I can say one thing the model cannot show.

Choose two stations. Give a claim and matching clue for each. Name one limit.

Talk first. You can plan in any language.

Point, sign, speak, draw, type, write, use a scribe or show a staff-captured photo.

Compare two models, clues and limits.

Say first: The ___ model fits because ___, but it cannot show ___.

Go deeper: Compare two models: what can each show and miss?

Model limits: Two toy runs differ on one surface. Can friction alone explain this? Choose one control that would help.

Question or model	My claim and clue	One limit
_____	_____	_____
_____	_____	_____

Choose one route within the full 8 minutes. Keep your own response.

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Practical model · safe steps and record

I can match a question to a model.

I can give a claim with a matching clue.

I can say one thing the model cannot show.

Guess first: Which surface may let the same toy travel farther?

Kit: Small toy · Rough and smooth paper · Start line · Ruler · Torch and ball · Warming cards

Mark one start and use the same toy.

Launch twice on each surface in the same way.

Measure each run and record it.

Use other models only for their own questions.

Risk: A moving toy may hit someone or fall from the table.

Safe: Use a low clear track. Keep hands and faces away from the run.

Surface	run 1	run 2
Rough	___	___
Smooth	___	___

Distance unit: _____. Use the same unit for all four runs.

Use the kit your teacher has checked. A picture or adult-led model is a valid route.

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Review and exit

I can match a question to a model.

I can give a claim with a matching clue.

I can say one thing the model cannot show.

Does your clue fit? Fix one clue or add a limit.

VF: hear one comment. NS(+): name a next step. //: edit one part. ?: ask for a read-back. Take one quiet minute.

Name a model, claim and clue. Say what changed after feedback.

? Read it back: choose a model that fits a science question and use a matching clue.

My feedback was… say what changed. Next I need to… name a fix.

How I know: I can match a question to a model. I can give a claim with a matching clue. I can say one thing the model cannot show.

The adult hears you and chooses our next example. Take one quiet minute.

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